

StudyForge

Pages

Comprehensive Exam Preparation Guide

C8 — The Periodic Table

1. Arrangement of Elements (C8.01)

Groups and Periods

The modern **Periodic Table** is based on the table proposed by Mendeleev in 1869. It is used to classify elements and predict their properties.

In the Periodic Table:

- Elements are arranged in order of increasing **proton (atomic) number**.
- Vertical columns of elements with similar properties are called **groups** (I to VIII).
- Horizontal rows are called **periods** (1 to 7).

Major regions of the Periodic Table:

Region	Description	Examples
Group I	Alkali metals — very reactive metals	Li, Na, K
Group II	Alkaline earth metals	Be, Mg, Ca
Transition elements	Hard, strong, dense metals in the centre block	Fe, Cu, Cr, Ni
Group VII	Halogens — reactive non-metals	F, Cl, Br, I
Group VIII	Noble gases — very unreactive	He, Ne, Ar
Metalloids	Semiconductors along the metal/non-metal boundary	Si, Ge

Metals and Non-metals

- **Metals** are found on the **left** and **centre** of the Periodic Table.
- **Non-metals** are grouped in the **top right-hand region**, above the thick stepped line.
- The boundary is not sharp — elements near the line are called **metalloids (semi-metals)**.

Metalloids have properties of both metals and non-metals:

- They look like metals but are **brittle** like non-metals.
- They are neither good conductors nor good insulators — they make excellent **semiconductors**.
- The prime example is **silicon (Si)**, the basis of the semiconductor industry.
- There are **eight** metalloid elements.

Position of Hydrogen

Hydrogen is difficult to place in the Periodic Table:

- It is sometimes placed above Group I or Group VII.
- In modern tables it is often shown **on its own**, separate from any group.
- As the smallest atom, its properties are **unique** — it does not fit neatly into the trends of any single group.

Electronic Configuration and the Periodic Table

There is a direct link between an element's position in the Periodic Table and its **electron arrangement**:

- **Group number** = number of electrons in the **outer shell** (for main-group elements).
- **Period number** = number of **electron shells** the atom has.

- Each period represents the **filling of an electron shell**.

Examples:

Element	Group	Period	Electron Configuration
Lithium (Li)	I	2	2,1
Beryllium (Be)	II	2	2,2
Carbon (C)	IV	2	2,4
Nitrogen (N)	V	2	2,5
Neon (Ne)	VIII	2	2,8
Sodium (Na)	I	3	2,8,1
Magnesium (Mg)	II	3	2,8,2
Silicon (Si)	IV	3	2,8,4
Chlorine (Cl)	VII	3	2,8,7
Argon (Ar)	VIII	3	2,8,8
Potassium (K)	I	4	2,8,8,1
Calcium (Ca)	II	4	2,8,8,2

Why elements in the same group have similar properties:

- They have the **same number of outer-shell electrons**.
- It is the **outer electrons** that are mainly responsible for an element's chemical properties.
- The most stable electron configurations have **full outer shells** (like Group VIII) — this explains why noble gases are so unreactive.

General Trends Across the Periodic Table

Across a period (left to right):

- Elements change from **metallic** to **non-metallic** character.
- Atoms get **smaller** and **less metallic**.

Down a group:

- Metals get **more reactive** (easier to lose outer electrons).
- Non-metals get **less reactive** (harder to gain electrons).
- Atoms get **larger** and **more metallic**.
- **Densities and melting points increase** down any group (general trend).

Key facts:

- Caesium (Cs) is the **most reactive metal** available in useful amounts.
- Fluorine (F) is the **most reactive non-metal**.

2. Group I — The Alkali Metals (C8.02)

Physical Properties

The metals in Group I are called the **alkali metals**: lithium (Li), sodium (Na), potassium (K), rubidium (Rb), caesium (Cs).

General physical properties:

- They are **soft solids** that can be cut with a knife.

- They have relatively **low melting points** and **low densities**.
- They are stored in **oil** to prevent reaction with oxygen and water vapour.
- When freshly cut, they have a **light-grey, silvery surface** that quickly tarnishes and becomes dull.

Physical properties data:

Metal	Electron Config.	Density (g/cm ³)	Melting Point (°C)	Boiling Point (°C)	Hardness
Lithium	2,1	0.53	181	1342	Fairly soft
Sodium	2,8,1	0.97	98	883	Soft
Potassium	2,8,8,1	0.86*	63	760	Very soft

Note: The densities of rubidium and caesium are 1.53 and 1.88 g/cm³ respectively.

Trends down Group I (physical):

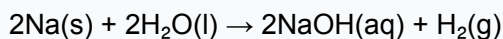
- Melting points and boiling points **decrease**.
- Metals get **softer**.
- Densities generally **increase** (sodium is anomalously high).

Chemical Properties — Reaction with Water

All Group I metals react with cold water to form **hydrogen** and an **alkaline solution** of the metal hydroxide. This is why they are called the **alkali metals**.

alkali metal + water → metal hydroxide + hydrogen

Example — sodium with water:



Trend: Chemical reactivity increases down the group:

Alkali Metal	Gas Production	Heat Produced	Alkaline Solution	Flame
Lithium	Fizzes slowly, few bubbles	Moves slowly on surface, does not melt	Metal disappears slowly, turns purple	No flame
Sodium	Fizzes strongly, many bubbles	Melts into a ball, moves about	Metal disappears quickly, turns purple	No flame (unless held in place)
Potassium	Fizzes violently, very many bubbles	Melts and moves very quickly	Disappears very quickly, turns purple	Hydrogen ignites, lilac flame

Why reactivity increases down the group:

- As you go down the group, the outer electron is **further from the nucleus** and more **shielded** by inner electron shells.
- This means the outer electron is **easier to lose**, so the metal is more reactive.

3. Group VII — The Halogens (C8.03)

Physical Properties

The most reactive non-metals are the **halogens**, found in Group VII. They exist as **diatomic molecules** (pairs of atoms): Cl₂, Br₂, I₂.

Common properties of all halogens:

- They are all **poisonous** and have a similar strong smell.
- They are all **non-metals**.
- They all form **diatomic molecules** (e.g. Cl₂, Br₂, I₂).
- They all have a **valency of 1** and form compounds called **halides** (e.g. HCl, HBr, HI).

Physical properties data:

Halogen	Electron Config.	Melting Point (°C)	Boiling Point (°C)	State at r.t.p.	Colour
Chlorine	2,8,7	-101	-35	Gas	Pale yellow-green
Bromine	2,8,18,7	-7	+59	Liquid	Red-brown
Iodine	2,8,18,18,7	+114	+184	Solid	Grey-black

Trends down Group VII (physical):

- Melting points and boiling points **increase**.
- Densities **increase**.
- Physical state changes from **gas** → **liquid** → **solid**.
- Colour intensifies from **pale to dark**.

Chemical Properties — Reactivity Decreases Down the Group

In contrast to Group I, **reactivity decreases** as you go down Group VII:

- Fluorine is the **most reactive** halogen (and most reactive non-metal of all).
- Iodine is the **least reactive** of the common halogens.

Why reactivity decreases down the group:

- As you go down the group, the outer shell is **further from the nucleus** and more **shielded**.
- This makes it **harder to attract** an extra electron to complete the outer shell.

Halogen Displacement Reactions

A **more reactive halogen** will **displace** a less reactive halogen from a solution of its halide salt.

Reactivity order: Chlorine > Bromine > Iodine

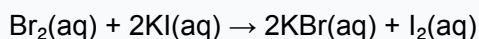
Displacement reactions:



Chlorine displaces bromine — solution turns **orange** (from colourless).



Chlorine displaces iodine — solution turns **yellow-brown**.



Bromine displaces iodine.

Displacement summary table:

Reaction with	Chlorine	Bromine	Iodine
Chlorides	—	No reaction	No reaction

Bromides	Displaces Br ₂	—	No reaction
Iodides	Displaces I ₂	Displaces I ₂	—

Rule: A halogen can only displace a halogen **below it** in Group VII (a less reactive one).

4. Transition Elements (C8.04)

The **transition elements** (or transition metals) form a large block in the **centre** of the Periodic Table, between Group II and Group III (in Period 4 and beyond).

General Physical Properties

Compared to Group I metals, transition metals are:

- **Hard and strong** (not soft like alkali metals).
- **High density**.
- **High melting and boiling points** (e.g. tungsten melts at 3410 °C — used in light bulb filaments).
- Much **less reactive** than Group I and II metals.
- Many have excellent **corrosion resistance** (e.g. chromium).

Distinctive Properties

1. **Variable oxidation states** — transition metals can form ions with different charges (e.g. Fe²⁺ and Fe³⁺, Cu⁺ and Cu²⁺).
1. **Coloured compounds** — many transition metal compounds are coloured:
 - Copper compounds: blue/green
 - Iron(II) compounds: green
 - Iron(III) compounds: orange/brown
 - Chromium compounds: green/orange
 - Nickel compounds: green
1. **Catalytic activity** — transition metals and their compounds often act as **catalysts**:
 - **Iron** — catalyst in the Haber process (making ammonia).
 - **Platinum** and **rhodium** — catalysts in catalytic converters.
 - **Nickel** — catalyst in hydrogenation of vegetable oils.

Common transition metals and their uses:

Metal	Uses
Iron (Fe)	Nails, bridges, machinery; catalyst (Haber process)
Copper (Cu)	Electrical wiring, water pipes, coins
Chromium (Cr)	Chrome plating (corrosion resistance)
Titanium (Ti)	Camera bodies, aircraft, medical implants
Tungsten (W)	Light bulb filaments (very high melting point)
Gold (Au)	Jewellery, electronics connectors
Silver (Ag)	Cutlery, photography, electronics
Platinum (Pt)	Catalytic converters, jewellery

5. The Noble Gases — Group VIII (C8.05)

The **noble gases** are the elements in Group VIII (also called Group 0): helium (He), neon (Ne), argon (Ar), krypton (Kr), xenon (Xe), radon (Rn).

Key Properties

- They are **very unreactive** (chemically inert).
- They exist as **monatomic gases** (single atoms, not molecules).
- They make up about **1%** of the Earth's atmosphere (argon is the most common).
- Their atoms have a **complete outer shell of electrons** (8 electrons, except helium which has 2) — this is an energetically **stable** arrangement.

Why they are unreactive:

- A full outer shell means the atom has no tendency to gain, lose, or share electrons.
- They do not combine with other atoms to form molecules or any other structure.

Discovery: Argon was discovered in 1894 by Lord Rayleigh. Helium was first detected by spectroscopy of sunlight during an eclipse. Other noble gases were isolated by Sir William Ramsay in the 1890s. Radon (radioactive) was the last to be purified (1909).

Uses of Noble Gases

Noble Gas	Uses
Helium	Weather balloons, airships (lighter than air, non-flammable)
Neon	Advertising signs (glows red-orange in discharge tubes)
Argon	Filling light bulbs (inert atmosphere prevents filament oxidation)
Krypton	Photographic flash equipment

6. Key Terms

Term	Definition
Periodic Table	A table of elements arranged in order of increasing atomic number, with elements of similar properties in vertical columns (groups)
Group	A vertical column in the Periodic Table; elements in the same group have the same number of outer electrons and similar chemical properties
Period	A horizontal row in the Periodic Table; the period number equals the number of electron shells
Alkali metals	The highly reactive metals in Group I (Li, Na, K, Rb, Cs); they react with water to form alkaline solutions
Halogens	The reactive non-metals in Group VII (F, Cl, Br, I); they exist as diatomic molecules
Halides	Compounds formed by halogens with metals or hydrogen (e.g. NaCl, HBr, KI)
Noble gases	The very unreactive elements in Group VIII (He, Ne, Ar, Kr, Xe, Rn); they have complete outer electron shells
Transition elements	The block of metals in the centre of the Periodic Table; they are hard, dense, have high melting points, form coloured compounds, and act as catalysts
Metalloid (semi-metal)	An element with properties between metals and non-metals; they are semiconductors (e.g. silicon, germanium)

Displacement reaction	A reaction where a more reactive element replaces a less reactive element in a compound
Diatomic molecule	A molecule consisting of two atoms bonded together (e.g. Cl ₂ , Br ₂ , I ₂ , H ₂ , O ₂)
Monatomic	Existing as single atoms (not bonded into molecules); noble gases are monatomic
Periodic property	A property that shows a repeating pattern across the periods of the Periodic Table
Electronic configuration	The arrangement of electrons in shells around the nucleus of an atom (e.g. 2,8,1 for sodium)
Catalyst	A substance that speeds up a chemical reaction without being used up itself
Valency	The combining power of an atom; halogens have a valency of 1

7. Key Equations

Group I — Alkali Metals with Water

- $2\text{Li(s)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{LiOH(aq)} + \text{H}_2\text{(g)}$
- $2\text{Na(s)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{NaOH(aq)} + \text{H}_2\text{(g)}$
- $2\text{K(s)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{KOH(aq)} + \text{H}_2\text{(g)}$

General Pattern

- alkali metal + water → metal hydroxide + hydrogen

Halogen Displacement Reactions

- $\text{Cl}_2\text{(aq)} + 2\text{KBr(aq)} \rightarrow 2\text{KCl(aq)} + \text{Br}_2\text{(aq)}$
- $\text{Cl}_2\text{(aq)} + 2\text{KI(aq)} \rightarrow 2\text{KCl(aq)} + \text{I}_2\text{(aq)}$
- $\text{Br}_2\text{(aq)} + 2\text{KI(aq)} \rightarrow 2\text{KBr(aq)} + \text{I}_2\text{(aq)}$

8. Practice Questions with Full Answers

Question C8.01

What is the name of the alkali formed when potassium reacts with water?

Answer: Potassium hydroxide (KOH).

Question C8.02

Write a word equation for the reaction between lithium and water.

Answer: lithium + water → lithium hydroxide + hydrogen

Question C8.03

Using the trends in Tables C8.01 and C8.02: (a) Suggest a value for the melting point of rubidium. (b) Suggest how the reactivity of rubidium with water compares with potassium.

Answer: (a) The melting points decrease down Group I: Li = 181 °C, Na = 98 °C, K = 63 °C. The gap is getting smaller each step (~83, then ~35). Rubidium's melting point would be about 20–30 °C lower than potassium, so approximately 33–43 °C. The actual value is 39 °C. (b) Rubidium is below potassium in Group I, so it will be more reactive. It will react even more vigorously with water than potassium — likely explosive, with an immediate flame.

Question C8.04

Which halogen(s) will displace bromine from a solution of potassium bromide?

Answer: Only **chlorine** (and fluorine) will displace bromine, because they are more reactive. Iodine will NOT displace bromine because it is less reactive. $\text{Cl}_2(\text{aq}) + 2\text{KBr}(\text{aq}) \rightarrow 2\text{KCl}(\text{aq}) + \text{Br}_2(\text{aq})$.

Question C8.05

What is the similarity in the electron arrangement of the noble gases?

Answer: All noble gases have a **complete (full) outer shell** of electrons. Helium has 2 electrons in its outer shell; all others (Ne, Ar, Kr, Xe, Rn) have 8 electrons in their outer shell. This stable configuration is why they are unreactive.

Question C8.06

State four differences between an alkali metal and a transition metal such as nickel.

Answer: (1) Alkali metals are soft; transition metals are hard and strong. (2) Alkali metals have low density; transition metals have high density. (3) Alkali metals have low melting points; transition metals have high melting points. (4) Alkali metals are very reactive (react with water); transition metals are much less reactive.

Question C8.07

Which phrase completes: "All transition metals in their compounds..." (A) are white in colour, (B) are magnetic, (C) form ions of the type M^{2+} and M^{3+} , (D) often act as catalysts.

Answer: (C) form ions of the type M^{2+} and M^{3+} — this describes their variable oxidation states. Note: (D) is also a distinctive property of transition metals, but the question says "all ... in their compounds" — variable oxidation states is the most accurate general statement about their compounds.

Question C8.08

Explain why elements in the same group of the Periodic Table have similar chemical properties.

Answer: Elements in the same group have the **same number of electrons in their outer shell**. Since it is the outer-shell electrons that are mainly responsible for chemical reactions (bonding, losing or gaining electrons), elements with the same outer-shell configuration react in similar ways.

Question C8.09

Describe the trend in reactivity going down (a) Group I and (b) Group VII. Explain each trend.

Answer: (a) In Group I, reactivity **increases** down the group. As atoms get larger, the outer electron is further from the nucleus and more shielded by inner shells, making it easier to lose — so the metal reacts more vigorously. (b) In Group VII, reactivity **decreases** down the group. As atoms get larger, the outer shell is further from the nucleus, making it harder to attract an extra electron to complete the shell — so the non-metal reacts less vigorously.

Question C8.10

The figure shows an outline of the Periodic Table with elements V, W, X, Y marked. V is in Group I Period 4, Y is in Group II Period 4, W is a transition element, X is in Group VII Period 3. Identify which combination is a metal and a non-metal.

Answer: X is in Group VII (a non-metal/halogen) and Y is in Group II (a metal/alkaline earth). So the correct answer is: Non-metal = X, Metal = Y. V is also a metal (Group I). W is a transition metal.

Question C8.11

Identify which combination shows the order of increasing reactivity in Group I and Group VII.

Answer: Group I: reactivity increases going down, so $\text{Li} \rightarrow \text{Cs}$ (least to most reactive). Group VII: reactivity increases going up, so $\text{I} \rightarrow \text{F}$ (least to most reactive). The correct combination is: Group I = $\text{Li} \rightarrow \text{Cs}$, Group VII = $\text{I} \rightarrow \text{F}$ (Answer B).

Question C8.12

From a table of Group I and Group VII elements (Li, Na, K, Cl, Br, I), identify: (a) a solid Group VII element, (b) the Group I element with the lowest density, (c) an element that is a liquid at room temperature, (d) the two elements that would react most violently with each other.

Answer: (a) Iodine — it is a grey-black solid at room temperature. (b) Lithium — density 0.53 g/cm^3 (it actually floats on water). (c) Bromine — it is a red-brown liquid at room temperature. (d) Potassium and chlorine — potassium is the most reactive Group I element listed, and chlorine is the most reactive Group VII element listed, so they would react most violently together.

Question C8.13

(a) State whether transition elements have high or low densities. (b) State whether transition elements are all gases or all metals. (c) Give the name of the group that includes helium, neon, and argon.

Answer: (a) Transition elements have **high** densities. (b) They are **all metals**. (c) The **noble gases** (Group VIII / Group 0).

Question C8.14

Describe the change in character of elements across a period from left to right.

Answer: Elements change from **metallic** character on the left to **non-metallic** character on the right. The elements on the far left (Group I) are highly reactive metals, those in the middle are transition metals, the metalloids are near the boundary, and the elements on the far right (Group VII) are reactive non-metals, with Group VIII being unreactive noble gases.

Question C8.15

Explain why chlorine can displace bromine from potassium bromide solution but iodine cannot.

Answer: Chlorine is **more reactive** than bromine (higher in Group VII, so its atoms can more easily attract an extra electron). Therefore chlorine can displace the less reactive bromine: $\text{Cl}_2 + 2\text{KBr} \rightarrow 2\text{KCl} + \text{Br}_2$. Iodine is **less reactive** than bromine (lower in Group VII), so it cannot displace bromine — a less reactive halogen cannot displace a more reactive one.

Question C8.16

Predict two properties you would expect the element rubidium (Group I, Period 5) to have, based on its position in the Periodic Table.

Answer: (1) Rubidium will be a very soft metal that can be cut with a knife (softer than potassium). (2) Rubidium will react very vigorously/explosively with water to form rubidium hydroxide and hydrogen, more vigorously than potassium (reactivity increases down Group I).

Question C8.17

Name three distinctive properties of transition metals that are NOT shared by Group I metals.

Answer: (1) Transition metals form **coloured compounds** (e.g. blue copper sulfate, green iron(II) chloride). (2) Transition metals have **variable oxidation states** (e.g. Fe^{2+} and Fe^{3+}). (3) Transition metals and their compounds often act as **catalysts** (e.g. iron in the Haber process, platinum in catalytic converters).

Question C8.18

Explain why the noble gases are unreactive.

Answer: Noble gases have a **complete outer shell of electrons** (2 for helium, 8 for all others). This arrangement is energetically very stable. Because the outer shell is full, noble gas atoms have no tendency to gain, lose, or share electrons, so they do not form bonds or take part in chemical reactions.

9. Exam Focus Priorities (Ranked by Frequency)

1. **Group I trends and reactions** — reactivity increases down the group; reaction with water produces metal hydroxide + hydrogen; linking trends to atomic structure (outer electron further from nucleus).
 2. **Group VII trends and displacement** — reactivity decreases down the group; halogen displacement reactions with halide solutions; writing equations.
 3. **Electronic configuration and position** — using group number (= outer electrons) and period number (= shells) to predict or identify an element's position.
 4. **Comparing Group I vs transition metals** — listing differences in physical properties (hardness, density, melting point, reactivity) and distinctive transition metal properties (coloured compounds, variable oxidation states, catalysts).
 5. **Noble gases** — explaining unreactivity in terms of full outer electron shells; monatomic nature.
 6. **Predicting properties** — using trends within a group to predict melting points, reactivity, or physical state of elements not directly studied (e.g. rubidium, astatine).
-

10. Quick Revision Card

- **Periodic Table:** elements arranged by increasing atomic number; **groups** = vertical columns (similar properties); **periods** = horizontal rows.
- **Group number = outer electrons; Period number = number of shells.**
- Elements change from **metallic** → **non-metallic** across a period.
- **Group I (alkali metals):** soft, low density, low m.p., very reactive with water → $\text{MOH} + \text{H}_2$. Reactivity **increases** down the group.
- **Group VII (halogens):** diatomic (Cl_2 , Br_2 , I_2), poisonous, coloured. Reactivity **decreases** down the group. More reactive halogen displaces less reactive: $\text{Cl}_2 + 2\text{KBr} \rightarrow 2\text{KCl} + \text{Br}_2$.
- **Transition metals:** hard, dense, high m.p., coloured compounds, variable oxidation states, catalysts.
- **Noble gases (Group VIII):** unreactive, monatomic, full outer shell of electrons.
- **Metalloids:** semiconductors at the metal/non-metal boundary (Si, Ge).